SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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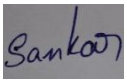
COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct

I Kadiri Sankar Narayana(192325085)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks - Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

   A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

1. Cloud Infrastructure

- Cloud servers host the inventory management application.
- Cloud storage stores product details, sales records, and transaction logs.
- Virtual machines provide computing resources for the system.

2. Client Systems

- POS (Point of Sale) systems update inventory after every sale.
- Web and mobile dashboards allow managers to monitor stock levels.
- Store employees can access inventory information from any branch.

3. Cloud Services

- Platform services analyze sales data and predict future demand.
- APIs enable communication between POS systems and cloud servers.
- Analytics services generate reports and business insights.

4. Data Management

- Product information is stored in cloud databases.
- Transaction logs are securely maintained for auditing.
- Analytics datasets help identify sales trends and customer preferences.

5. Security

- VPNs provide secure communication between stores and the cloud.
- Firewalls protect the system from unauthorized access.
- API gateways authenticate and manage all API requests.

6. Benefits

- Real-time inventory updates across all stores.
- Improved demand forecasting and stock management.
- Secure, scalable, and cost-effective cloud solution.
- Better decision-making through analytics and centralized data management.

Case Study 2: Cloud-Based Document Collaboration System

1. Cloud Platform

- The document collaboration application is hosted on the cloud.
- Users can access their documents anytime using a web browser.
- The cloud automatically handles storage and system maintenance.

2. User Access

- Multiple users can edit the same document at the same time.
- Changes made by one user are instantly visible to others.
- The system can be accessed from laptops, tablets, and mobile phones.

3. Document Storage

- Documents are stored in distributed cloud storage for better reliability.
- Version control keeps track of all changes and allows users to restore older versions if needed.
- Data is backed up regularly to prevent data loss.

4. Synchronization

- Web APIs keep documents synchronized across different devices.
- Any edits are updated in real time without needing to refresh the page.
- This ensures everyone always works on the latest version.

5. Network Performance

- A global cloud network provides fast access with low latency.
- High availability ensures the service remains accessible even if one server fails.
- Load balancing helps maintain smooth performance during heavy usage.

6. Security

- Documents are encrypted while being stored and transferred.
- Identity management verifies users before granting access.
- Access permissions allow document owners to control who can view, edit, or share files.

Case Study 3: Online Food Ordering Platform

1. Cloud Infrastructure

- The food ordering application is hosted on a cloud platform.
- Virtual machines and managed Kubernetes help the system handle a large number of users.
- The infrastructure can scale automatically during peak ordering hours.

2. Client Applications

- Customers can place orders using a web browser or a mobile app.
- Restaurant staff can receive and manage orders through the platform.
- Users can track their orders in real time.

3. Backend Services

- RESTful Web APIs connect the frontend applications with the backend services.
- APIs process customer orders, payments, and delivery updates.
- The system exchanges data quickly between users and restaurants.

4. Cloud Storage

- Menu images, customer details, and invoices are stored securely in cloud storage.
- Order history is maintained for future reference.
- Regular backups help protect important business data.

5. Scalability and Reliability

- Managed Kubernetes automatically manages application deployment and scaling.
- Virtual machines provide the required computing resources.
- The platform remains available even during busy hours with many users.

6. Security

- Authentication tokens verify the identity of users before they access the system.
- HTTPS encrypts data transferred between clients and servers.
- Role-based access control ensures that customers, restaurant staff, and administrators can only access the features assigned to them.